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Chapter 19: The Spatialization of the Mapping Chain—A Fiber Diagnosis of Global Continuous Trading and Cross-Market Pricing

Abstract

This chapter extends the fiber diagnosis of the financial mapping chain from the time window of a single market to the spatial dimension of the global network, analyzing in depth the structural reconstruction by which financial markets have evolved from "intermittent trading" to "continuous trading." Through empirical analysis of phenomena such as the night-session system, the proxy pricing of cross-border ETFs and ADRs, 24-hour continuous trading, the global financial cycle, and market fragmentation, this chapter reveals the mechanism by which pricing power is redistributed geographically and temporally. The analysis of this chapter shows that cross-market and cross-time-zone continuous trading turns the mapping chain from a single-market chain into a global network: the range of access to independent information widens, while mutual citation among readings intensifies; when arbitrage, data integration, and regulatory coordination fail to adjust in step, the uncertainties of proxy pricing and cross-venue integration may be concealed beneath precise readings, giving rise to the "pseudo-small fiber of proxy pricing" and the continuous inflation of hidden variables. At the same time, the algorithmization of high-frequency trading and of audit tools not only changes market microstructure but also completes a feedback loop into the real economy through the channel of the cost of capital. This chapter argues that in global continuous markets no single reading suffices to uniquely restore the true state, and that the spatialization of the mapping chain requires us to re-examine financial audit and price discovery at the epistemological

level.

Keywords

continuous trading; spatialization of the mapping chain; cross-border pricing; pseudo-small fiber; market fragmentation; global financial cycle

§1 Introduction: From Speed to Continuity

Chapter 17 established, within the time window of a single market, the skeletal fiber diagnosis of the financial mapping chain: the four-layer diagnosis of L0 mapping-chain structure, L1 optimization dynamics, L2 pseudo-calibration, and L3 multi-mapping recursion (this is the four-layer progressive structure of Chapter 4, §7 as applied at the scale of a single market; the finance section's five-layer progressive structure additionally contains L4 incentive capture—see Chapter 17, §0—to be taken up by subsequent chapters). The object of diagnosis in this chapter shifts to another layer of structural change: the transformation of global financial markets from "intermittent trading" to "continuous trading." It extends the fiber diagnosis of Chapter 17 to the cross-market, cross-time-zone network, and examines how global 24-hour continuous trading carries the reticulation of the mapping chain. The diagnosis along the speed dimension has already been carried out by Chapter 7 as an extreme empirical case—high-frequency trading response times as low as 42 nanoseconds (MacKenzie, 2021), against a human reaction time of about 200 milliseconds, a rate difference of about five million times—so this chapter will not repeat it; this chapter focuses on the structural change from intermittent to continuous trading, and §7 will complete the feedback loop from market microstructure to the real economy.

Over the past thirty years, global financial markets have undergone a profound structural evolution: from "intermittent trading" to "continuous trading." This transformation was not a sudden leap from nothing to something. The foreign-exchange market had long already been a 24-hour continuous market, as Glantz & Kissell (2014) note, it is "the only truly continuous and nonstop trading market in the world, with participants trading day and night, weekday and weekend, and on holidays" (following the wording of Glantz and Kissell's original text; translator's note: strictly speaking, contemporary mainstream foreign-exchange markets run on a 24/5 basis, continuously from Sunday through Friday, with major trading centers suspending market-making each weekend and on major holidays; the original phrase "trade on weekends and holidays" is an exaggerated generalization, since only offshore brokers offer limited quotes within the very thin weekend trading window), and multinational banks had long already been making markets across time zones. The crucial structural change is that continuity expanded from a marginal practice of the foreign-exchange and derivatives markets into an institutional characteristic of mainstream asset classes, from the exclusive preserve of large institutions to broad retail access, and from a "supplementary session" reviewed after the trading day into an independent "pricing node" that actively drives price discovery. The long history of financial globalization, the collapse of Bretton Woods, the opening of capital accounts, and the integration of cross-border payment systems, supplied the institutional background for this transition. But what is specific to the last thirty years is that continuity has penetrated from the pricing of currencies and interest rates into the pricing of equities and commodities (Caprio, 2013). Night-session systems allow local markets to continue operating beyond regular hours. Cross-border exchange-traded funds (ETFs) and American Depositary Receipts (ADRs) allow investors to engage in proxy pricing during the closure of the underlying market. Twenty-four-hour futures trading disperses price discovery across multiple time zones. The synchronized driving force of the global financial cycle causes multiple markets

to move in the same direction even without direct causal interaction.

In the language of the Fiber Framework, this transition means a fundamental reconstruction of the mapping chain. The single-market mapping chain analyzed in Chapter 17 can, setting aside the intra-market citation of readings, be simplified into a linear chain: $T \to r_1 \to r_2 \to \cdots$; its L3 layer already contains recursion, and this chapter extends it to the cross-market network. But when price discovery changes from a single market session to a global 24-hour continuous relay, the mapping chain becomes reticular: the same true state T is simultaneously captured by dispersed readings from multiple markets, these readings calibrate one another and cite one another, and a "global price" finally emerges. Given any single reading, the partial inputs of its pricing function are no longer confined to a single market. Cross-market and cross-time-zone mutual citation reshapes the structure of the partial input network: newly added independent readings may shrink the set of compatible states, while repeated citation of existing readings need not add new constraints—when the citation chain grows faster than the pool of independent calibration sources, a single reading accommodates more compatible interpretations and the fiber tends to inflate. What determines pricing is changing: "whose model is more accurate" is no longer the only answer; one must also ask, "whose market happened to be open when the information shock occurred?" This judgment will be gradually unfolded in the empirical analyses of §§2-4.

Behind these changes there is one deeper shared structural feature: all of them alter the **input composition** of the "partial input network" of financial readings, the concept defined in Chapter 17, §1.2. The night session widens the hours during which local information can enter prices; cross-border proxy pricing introduces valuation time lags and cross-market citation; global common shocks may lower the relative share of local idiosyncratic information—the three alter the input structure in different ways. The concrete mechanisms will be unfolded section by section from §2 to §7, and synthesized in §8.

This chapter unfolds the diagnosis according to the following structure: §§2-4 analyze respectively how the night-session system, cross-border ETFs/ADRs, and extended trading hours reconstruct the temporal and spatial structure of the mapping chain; §5 introduces the global financial cycle and analyzes synchronized pricing under common driving factors; §6 diagnoses the data-integration dilemma produced by market fragmentation; §7 traces how high-frequency trading transmits market microstructure into the real economy through the channel of the cost of capital, completing the feedback loop; and §8 provides an integrated diagnosis through the four-layer progressive structure.

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§2 The Night-Session System and the Geographic Transfer of Pricing Power

§2.1 The Price-Discovery Function of After-Hours Trading

The classic study by Barclay and Hendershott in the *Review of Financial Studies* systematically analyzed the informational content of after-hours trading (Barclay & Hendershott, 2003). Their core finding is that although after-hours trading volume is far smaller than that of the regular session, after-hours trading contains significant price-discovery information. Trading costs related to bid-ask spreads in after-hours trading are four to five times higher than during the daytime session, and yet the informational contribution of after-hours trading to price discovery is disproportionately large. This means that price discovery does not occur only during the "high-liquidity" daytime session; it proceeds continuously across the full twenty-four hours.

Even when trading volume is low and spreads are wide, the reading, the price, formed during the after-hours session still carries independent information about the true state (T). If the auditor focuses only on daytime trading data, the informational input of the after-hours session is systematically omitted. After-hours sessions may reveal information that has not yet entered daytime prices, but may also generate greater execution noise from thin liquidity; the net effect on identification depends on the relative strength of the two and cannot be judged from volume or spreads alone—yet the inputs of the after-hours mapping function contain independent informational dimensions absent from the daytime mapping function.

§2.2 The Introduction of Night Trading in China's Futures Markets

The study by Jiang, Kellard, and Liu in the *Journal of Futures Markets* provides direct evidence that the introduction of a night session changes market quality and the geographic distribution of pricing power (Jiang, Kellard & Liu, 2020). They find that after the Shanghai Futures Exchange (SHFE) introduced a night trading session for precious metals in July 2013, trading activity in the Chinese market (volume and open interest) rose significantly, liquidity improved (the Roll and Amihud liquidity measures declined), and the two-way volatility spillover with the Commodity Exchange (COMEX) strengthened (especially in the U.S.-to-China direction). Notably, the price-discovery share of gold futures actually declined in relative terms, which the authors diagnose as a sign of reduced market segmentation and improved information flow, not of deteriorating Chinese market quality. In other words, the core achievement of the night session was not "recapturing price-discovery power," but connecting the Chinese market into the COMEX-dominated 24-hour continuous mapping chain; Chinese investors became able to respond immediately in the domestic market to overseas information, and pricing power was partially redistributed along the cross-market dimension.

Klein and Todorova's study of night trading in aluminum and copper futures on the Shanghai Futures Exchange reveals the cross-variety differences of this pattern (Klein & Todorova, 2021): after the introduction of night trading, the realized volatility of copper's night session is driven mainly by the volatility of the prior trading session on the London Metal Exchange (LME), exhibiting immediate linkage with international markets; aluminum, by contrast, is less sensitive to external shocks and displays pronounced long memory, and has not formed linkage with the LME to the same degree. The two authors also note that the informational content of the day session and the night session differs significantly, and that high-frequency data must be treated separately for each.

This is a story about the "access right" of the mapping chain. Before the Chinese market had a night session, the mapping chain of global commodities during U.S. trading hours was complete, U.S. investors -> U.S. futures market -> price, but broken during the hours when the Chinese market was closed: most Chinese investors could not encode local information into the global price in real time. The introduction of the night-session system was, in essence, the connection of the Chinese mapping function, π, into the global network of 24-hour continuous pricing. The precondition of pricing power is not a question of "whose market is larger," but of "whose market happened to be open when the information shock occurred"—openness provides the condition of participation, while dominance also depends on liquidity, informational advantage, and arbitrage links. The institutional design of global continuous trading is, in reality, a redistribution of access rights to the mapping chain.

§2.3 Fiber Diagnosis: The Access-Right Essence of Pricing Power

The fiber significance of the night-session system can be summarized in three points. First, the length of the mapping chain depends not only on the number of layers of information processing, but also on the continuity of the time window. When a market happens to be closed at the time of an information shock, the mapping chain during that interval is broken, and local information cannot enter the global price. Second, the night-session system reconnects the broken mapping chain, making the local market a node in the global continuous mapping chain. Third, the geographic transfer of pricing power is, first of all, a redistribution of access rights to the mapping chain: only when the door is open can information be encoded into price through it.

Boundary marking: the sample period of Jiang et al. (2020) spans January 2008 to April 2016, and that of Klein & Todorova (2021) January 2010 to May 2018; their findings are based on observations from the years around the introduction of night trading and should not be extrapolated directly to all later periods. From November 2025 to January 2026, the main Shanghai silver contract rose from RMB 11,441 per kilogram to a peak of RMB 32,382 per kilogram (a gain of 183%), while over the same period the main COMEX silver contract rose 152%, with the domestic market clearly stronger (*Economic Observer*, 2026-01-31)—but the difference in gains is not itself an indicator of price-discovery share, nor can it separately identify the respective roles of the night-session system and the concurrent macro environment; price performance, speed of informational response, and contribution to price discovery must be measured separately.

Input-composition marking: the night-session system incorporates into the global pricing network Chinese local information on supply and demand that was previously not admitted. This input is independent of readings from other markets. It adds new geographic sources of independent components into the partial input network.

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§3 Cross-Border ETFs and ADRs: Proxy Pricing and Price Discovery in Closed Markets

§3.1 The Pricing Mechanism of International ETFs

2025–2026 cross-border ETF data snapshot: as of December 9, 2025, the total scale of Chinese cross-border ETFs reached RMB 939.087 billion, up 121% from RMB 424.220 billion at the beginning of the year (*Securities Daily*, Fang Lingchen, 2025-12-10); in January 2026, total scale exceeded RMB 1 trillion for the first time (Yicai, 2026-01-15). Alongside this expansion, premiums erupted in a concentrated wave: since December 2025, fourteen public-fund institutions have cumulatively issued more than 200 premium-risk warning announcements; on December 9, the Southern S&P 500 ETF stood at a premium of 3.41% over its indicative optimized portfolio value (IOPV) and the Boshi S&P 500 ETF at 5.58% (*Securities Daily*, Fang Lingchen, 2025-12-10); in March 2026, when geopolitical factors were boosting oil-and-gas theme performance, the crude-oil LOF of E Fund reached a premium of 41.01% (Cailian Press, 2026-03-25). By May 2026, the phenomenon had escalated into an industry norm: premium-risk warnings from public funds exceeded 3,500 over the preceding year (*Securities Times*, 2026-05-07), with the hot spots rotating to semiconductor themes, and the premium of the China–Korea Semiconductor ETF at one point exceeded 35% (*Shanghai Securities News*, 2026-05-18).

In fiber language, this is the recurrent appearance of the "pseudo-small fiber of proxy pricing"[1] across multi-

[1]The "pseudo-small fiber of proxy pricing" refers to situations in which a proxy price is taken as a sufficient identification of the underlying state while its valuation time lags, arbitrage constraints, or liquidity conditions are ignored—a specialized form of pseudo-small fiber (Chapter 4, §2.4) in cross-border / cross-time-zone proxy-pricing scenarios. Secondary-market trading of ETFs or ADRs is

ple periods—when primary-market subscription is restricted by tight qualified domestic institutional investor (QDII) quotas or funds' own size controls, the supply of new units is suppressed and subscription arbitrage can hardly compress the premium in time (whether redemption is likewise restricted depends on each fund's announcements), the secondary-market price then systematically detaches from the fund NAV that serves as the "anchor," and the precise price reading observed by investors ("this ETF is worth this price right now") conceals behind it a huge uncertainty ("this price has been bid up only because the product cannot be bought through the normal channel"). This causal chain, restricted subscription supply -> strengthened arbitrage constraints -> premium inflation, is based on wide consensus among market practitioners (*Economic Information Daily*, 2025-12-25; *Securities Daily*, Fang Lingchen, 2025-12-10; with quota tightness regarded as a key variable, *Securities Times*, 2026-05-07), but at present there is still a lack of strict identification of each factor's contribution.

Cross-border ETFs allow investors to engage in continuous pricing during the closure of the underlying market. When Tokyo is closed, a Japanese stock ETF listed in the United States still trades. Price changes in the ETF reflect the expectations of U.S. investors regarding the Japanese market; while Tokyo is closed the local market produces no new transaction prices—this does not mean that Japanese investors or information about Japanese firms cannot enter this pricing process through other channels. The ETF arbitrage mechanism—when ETF prices deviate from the value of the underlying securities, authorized participants can create or redeem ETF units to arbitrage the gap—should in theory anchor ETF prices to the NAV of the underlying portfolio. But in periods of market turbulence, arbitrage constraints cause ETF prices to deviate significantly from the underlying securities.

Within the Fiber Framework, cross-border ETFs introduce a new layer of mapping: true state, the fundamentals of Japanese firms -> Japanese local stock price, reading r_1 -> U.S. ETF price, reading r_2. When Tokyo is closed, r_1 stops updating, but r_2 continues to move. Changes in r_2 do not reflect new information about T (because the local price channel has suspended updating); what they reflect is the change in U.S. investors' expectations about T—a mapping that reflects U.S. market valuation and liquidity conditions more than calibration by local prices. The mapping chain is lengthened one layer further: when different combinations of fundamentals, risk appetite, and arbitrage constraints can all generate similar ETF prices, the price alone cannot distinguish these combinations—what expands here is the set of compatible interpretations of a single reading; and the relation between the readings, the ETF premium or discount itself, becomes a new source of uncertainty.

§3.2 Price Discovery through ADRs

American Depositary Receipts (ADRs) form another channel of cross-border pricing. Howe and Ragan (2002) find that ADRs have become the dominant market during periods when they trade simultaneously with the underlying asset. The center of price discovery shifts from the home market to the ADR market. The study of Spanish cross-listed stocks by Pascual and coauthors adds a qualification (Pascual, Pascual-Fuster & Climent, 2006): when the home market is developed and sufficiently liquid, price discovery is almost entirely dominated by the home market—Spanish Stock Exchange (denoted SSE in the paper) trading-related shocks explain 12% to 30% of the long-run variance, while the trading-related share on the NYSE side is indistinguishable from

itself real trading; "proxy" means that in the relevant settings they substitute for the underlying market in carrying the pricing function, and their price benchmark is the fund NAV (for ETFs) or the corresponding security price converted by depositary ratio and exchange rate (for ADRs). The broad sense also includes the similar structure found in the integration of multi-venue NBBO quotes under market fragmentation (§6)—the NBBO is a quote consolidation whose uncertainty structure resembles proxy pricing but is not the same trading mechanism. The two cases share one structure: a precise reading conceals the structural fact that "this reading is only a proxy, not a direct trade in the underlying market."

zero, with a negligible overall contribution. Grammig and coauthors, in their high-frequency study of three German stocks, DaimlerChrysler, Deutsche Telekom, and SAP, traded simultaneously on NYSE and XETRA, further characterize the concrete form of this bidirectional division of labor: the price discovery of the stock itself is accomplished mainly in the German home market, but exchange-rate shocks have stronger explanatory power for the price evolution on the NYSE side than for the German home price (Grammig, Melvin & Schlag, 2005). The home market mainly carries the "fundamental value" reading, and the ADR price is more sensitive to exchange-rate shocks: this is a difference in relative explanatory power, not an exclusive division of functions, and the two synthesize a cross-border price through mutual calibration. It should be noted that the empirical studies of Howe and Ragan (2002) and Grammig et al. (2005) are based on samples from the late 1990s to the early 2000s, and their conclusions presuppose that cross-market arbitrage can be conducted in a timely manner; when capital controls block the arbitrage channel, this premise does not hold.

Ma and McCauley (Ma & McCauley, 2013) provide a contrasting empirical study: in a sample of 9 triple-listed Chinese firms (Shanghai-Hong Kong-New York), over the period from March 1999 to June 2012, they use an autoregressive (AR) model to estimate the speed of mean reversion of the log price ratio $q_{i,t}$ between different market pairs (half-life $= -\ln 2/\ln(1+\beta)$): the half-life between the New York ADR and the Hong Kong H-share is **0.9 days**—under full openness to foreign investors and an open arbitrage channel, a spread shock decays by half within roughly one trading day, approaching the convergence speed under what the paper calls "instantaneous arbitrage" (a half-life measures the time for half-decay, not full convergence); between the New York ADR and the Shanghai A-share it is **114 days**; between the Hong Kong H-share and the Shanghai A-share it is **128 days** (reaching **233 days** in a wider sample of 41 A-H dual-listed firms); by way of comparison, the corresponding half-life for 9 New York-Mumbai dual-listed Indian firms is **48.6 days**. The paper accordingly notes that "policy continues to segment onshore and offshore markets in both and policymakers face challenges in further financial integration"—that is, policy still segments onshore and offshore markets in both economies; the speed of mean reversion of cross-market spreads is not a market-friction parameter, but a direct diagnostic signal of whether capital controls block the arbitrage channel. Concretely, for the same Hong Kong H-share, the half-life against the New York ADR is less than a day while the half-life against the Shanghai A-share is about 128 days: this contrast points to the constraint of capital controls, but the difference cannot be attributed to the QFII quota alone—share-conversion restrictions, investor eligibility, and other conditions coexist; the half-life reflects the strength of the constraints on cross-market links and cannot identify the net effect of any single control.

Within the Fiber Framework, this means that the direction of the mapping chain **depends on the institutional state of the arbitrage channel**, and presents two starkly different forms. When the arbitrage channel is open (as between the New York ADR and the Hong Kong H-share), the mapping chain is a bidirectional closed loop (ADR price $\leftrightarrow$ home-market price), and the two readings calibrate and influence one another; given either one of the two readings, the true state T cannot be uniquely determined, because the reading has itself already become a function of the other reading. This is the concrete form, in cross-market pricing, of the L3 layer of the Fiber Framework, multi-mapping recursion, meaning that multiple mapping functions form a closed loop of mutual citation such that the output of any single mapping function depends on the outputs of the others: the ADR market and the home market each become the other's "other clock," calibrating each other and forming a closure. But when the arbitrage channel is blocked by capital controls (as between New York/Hong Kong and Shanghai), multi-mapping recursion degenerates into slow correction under institutional damping, the L3 layer degenerates

from instantaneous calibration into asymptotic convergence—institutional constraints make spread correction slower and feedback lags longer, with buffer periods lasting months or even close to a year; the mapping chain is not broken at this point, nor does feedback disappear—it degenerates from instantaneous calibration into slow calibration with institutional damping: deviations are corrected with lags of months or longer, and persistent spreads can survive; this is a change in feedback strength and lag, which need not be renamed a "semi-closed loop." The speed of mean reversion of cross-market spreads thereby becomes an observable indicator of the discrete Leviathan's "regulatory tightness" on the cross-border capital channel.

§3.3 Fiber Diagnosis: The Pseudo-Small Fiber of Proxy Pricing

The pricing mechanisms of cross-border ETFs and ADRs create a particular form of pseudo-small fiber, δ_{pseudo}, meaning a state in which the reading appears precise while real uncertainty is concealed, the core concept of Chapter 4, §2.4. When Tokyo is closed, changes in the U.S. ETF price give investors a precise reading, "Japanese equities are now worth this price," but the uncertainty structure behind that reading is invisible to the investor: ETF prices may deviate from underlying securities because of liquidity shocks in the U.S. market; they may remain at sustained premiums or discounts because arbitrage is constrained; and they may be "corrected" by local prices once Tokyo opens. The precise number seen by the investor conceals the structural fact that "this is only proxy pricing, not a direct trade in the underlying market."

Boundary marking: the premium dilemma of cross-border ETFs reveals a key condition of the pseudo-small fiber of proxy pricing, the effectiveness of the arbitrage mechanism. When QDII quotas are abundant, authorized participants can create and redeem ETF units and pull prices back close to NAV, so the fiber is kept within a limited range by the arbitrage mechanism. But when quotas are tight or funds control their size and the subscription channel is restricted, arbitrage constraints strengthen markedly and the fiber tends to inflate. This means that the "pseudo-small fiber of proxy pricing" is not an inherent attribute of cross-border ETFs. It is **conditional on institutional circumstances**. When quotas are abundant, fiber is controllable; when subscription is restricted, fiber tends to inflate. The regulatory implication of this boundary is direct: the path to compressing the fiber is not to restrict cross-border ETFs as such, but to open the arbitrage channel, for example by enlarging QDII quotas and introducing more market makers.

Input-composition marking: pricing of cross-border ETFs during the closure of the underlying market is composed entirely of expectation. The reading produced by proxy pricing contains no independent information about the true state. It compresses the share of the independent component in the partial input network down to zero.

Spread persistence as an observable indicator of arbitrage constraints: the above dependence on boundary conditions is quantitatively contrasted in the China-Hong Kong cross-listed empirical study of §3.2: when the New York-Hong Kong arbitrage channel is fully open, the half-life is less than a day; when the New York-Shanghai or Hong Kong-Shanghai channel is blocked by capital controls, the half-life rises to between 114 and 233 days. The empirical study by Levy Yeyati, Schmukler, and Van Horen of 98 cross-listed stocks across nine emerging markets, 78 of which enter the threshold autoregression estimates (Argentina, Brazil, Chile, Indonesia, Korea, Mexico, Russia, South Africa, and Venezuela) (Levy Yeyati, Schmukler & Van Horen, 2009), using AR and threshold autoregressive (TAR) models to identify simultaneously the speed of convergence and the band of no-arbitrage (the threshold interval within which, when the cross-market spread falls inside it, arbitrage does not occur, because the deviation is not large enough to cover trading costs, foreign-exchange costs, and the implicit

tax of capital controls; only when the spread breaks out of the band does arbitrage force the deviation rapidly back inside), further provides a general conclusion: capital-outflow controls generate a positive premium (the home market is more expensive than the ADR), inflow controls generate a negative premium (the ADR is more expensive than the home market), and the size of the premium varies with the intensity of controls. Spread half-lives and bands of no-arbitrage provide observable indicators of arbitrage constraints; the chapter accordingly discusses the degree of constrained calibration, but does not equate these indicators with fiber size.

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§4 Twenty-Four-Hour Trading: The Trend from Margin to Mainstream

§4.1 Institutional Progress toward Extended Trading Hours

Between 2024 and 2026, 24-hour trading shifted from a "marginal experiment" to a "mainstream trend." On November 27, 2024, the SEC approved the Form 1 registration application of 24X National Exchange (Release No. 34-101777), the first time the SEC approved a national securities exchange operating on a nearly 24-hour basis. Previously, 24X's operator, 24 Exchange, had submitted to the SEC, in draft form in October 2021, an application for a national securities exchange trading seven days a week, 365 days a year; it withdrew the application on its own in June 2023 after running out of review time, and refiled in February 2024 under the name 24X National Exchange (Lyudvig, 2021; Bloomberg News, 2023; U.S. Securities and Exchange Commission, 2024). The 23-hour trading day of 24X consists of four sessions (ET): the Pre-Market Session (4:00-9:30 a.m.), the Core Market Session (9:30 a.m.-4:00 p.m.), the Post-Market Session (4:00-7:00 p.m.), and the overnight 24X Market Session (8:00 p.m. to 4:00 a.m. the next day), with a one-hour daily maintenance pause from 7:00-8:00 p.m.—these were the rules as approved in November 2024; in September 2025, 24X revised its rules to extend the Post-Market Session to 8:00 p.m., move the maintenance pause to 8:00-9:00 p.m., and change the overnight session to 9:00 p.m. to 4:00 a.m. the next day (U.S. Securities and Exchange Commission, 2025c). 24X is implementing this in stages: on October 14, 2025, the first three sessions (excluding the overnight session) were launched (4:00 a.m. to 8:00 p.m. ET), and the full 23/5 session is to begin after the corresponding amendments to the Equity Data Plans take effect (U.S. Securities and Exchange Commission, 2024, 2026).

In February 2025, the SEC approved by accelerated procedure the proposal to extend trading hours on NYSE Arca, a subsidiary of the New York Stock Exchange (Release No. 34-102400), extending NYSE Arca from its original three-session structure of 4:00 a.m.-8:00 p.m. ET (the Early Trading Session, the Core Trading Session, and the Late Trading Session) to 1:30 a.m.-11:30 p.m. ET Monday through Thursday and 1:30 a.m.-8:00 p.m. ET on Friday—a trading window of 22 hours Monday through Thursday and 18.5 hours on Friday (U.S. Securities and Exchange Commission, 2025a).

In 2025 Nasdaq announced a "24/5" roadmap (24 hours a day, 5 days a week), making clear that its landing depends on upgrades to exchange technology and supervisory capacity, an extended operating window for the Securities Information Processor (SIP), 24-hour clearing by the National Securities Clearing Corporation (NSCC) under the Depository Trust & Clearing Corporation (DTCC) (planned for completion in the second quarter of 2026, running continuously from 8:00 p.m. ET Sunday to 8:00 p.m. ET Friday), and the SEC's judgment on the scope of application across the night session of the Market Wide Circuit Breakers (MWCB), Limit Up/Limit Down (LULD), and Clearly Erroneous (CE) trade rules (Nasdaq, 2025).

In September 2025 the World Federation of Exchanges (WFE) released the policy report *Policy and Market*

Impacts of Extended Trading, defining extended trading as a near-around-the-clock capacity, typically 22 to 23 hours per day, 5 days per week, and systematically reviewing multidimensional challenges such as thin night-session liquidity with widening bid-ask spreads, the continued need to retain a daily closing or reference price for benchmarks and corporate actions, the need for clearing margins to adapt to a 24/7 data flow, and cross-border regulatory coordination (World Federation of Exchanges, 2025). Against this background, on January 8, 2026, One Trading announced that it had obtained an extended license from the Dutch Authority for the Financial Markets (Autoriteit Financiële Markten, AFM), describing its 24/7 equity perpetual futures market as "the first in regulated finance" (Finance Magnates, 2026).

Within the Fiber Framework, this series of institutional advances means that the time dimension of the mapping chain has been reconstructed. The traditional stock-market mapping chain is intermittent: it reconnects at every opening bell and disconnects at the close. Twenty-four-hour trading makes the mapping chain continuously connected: information shocks in more sessions gain the opportunity to enter price through trading; the speed of reflection still depends on market status, liquidity, and trading conditions—maintenance pauses and trading-day boundaries still exist. Fiber no longer stops updating during market closure; it changes continuously with the uninterrupted inflow of information. The challenge the WFE report emphasizes—that successful implementation of extended trading "requires a coordinated effort across the entire financial ecosystem", including regulatory frameworks—is, in fiber language, first of all a rise in the demands of regulatory division of labor and accountability; only when regulators also rely on one another's model outputs for calibration does the question "who audits the clock-setter?" further involve the L3 recursion.

§4.2 Asian Sessions and Overnight Pricing of U.S. Equities

Extended trading in U.S. equities and related products gives Asian investors more opportunities, within their local working hours, to respond to information published after the U.S. market close. What changes is the range of tradable sessions and participation, not the first appearance of off-hours pricing; whether a shift in the center of gravity of price discovery has occurred still requires comparing information contributions across sessions and cannot be inferred from the opening of a trading window alone.

Within the Fiber Framework, this means that the "input nodes" of the mapping chain have shifted from one to many. Previously, information released after the U.S. close was mostly encoded at the next New York opening (off-hours trading had long existed; see §2.1); extended sessions now allow Asian investors to actively encode information into price during their own local session. The input side of the mapping chain is no longer monopolized by New York, but dispersed across multiple time zones.

§4.3 Evidence of Scale: 24-Hour Trading in Foreign Exchange and Derivatives

BIS 2025 Triennial Survey provides evidence of scale. In April 2025, the global over-the-counter (OTC) foreign-exchange market reached an average daily trading volume of $9.6 trillion, up 28% from $7.5 trillion in 2022 (BIS, 2025a, notional amounts on a "net-net" basis); over the same period, average daily OTC interest-rate derivatives trading rose 59% from $5.0 trillion in April 2022 to $7.9 trillion (BIS, 2025b, notional amounts on a "net-net basis"). The foreign-exchange market is itself the most mature case of 24-hour continuous trading: it opens in Sydney, then passes successively through Tokyo, London, and New York, until New York closes and completes the full cycle.

Within the Fiber Framework, the 24-hour structure of foreign exchange is the institutionalized form of "relay pricing." The activity of trading centers shifts across intervals, and price formation takes on a relay character; but sessions overlap and participants can trade across regions, so the pricing power of any interval cannot be assigned exclusively to one geographic center. The price of the same currency pair in different intervals reflects the information and preferences of different market participants. Given a single price reading, one must know not only "which currency pair is this?" but also "in which interval was it formed?" because the set of true states compatible with the same reading, that is, the fiber, differs by interval.

§4.4 Fiber Diagnosis: The Dynamics of Inflation in the Continuous Mapping Chain

When the mapping chain changes from intermittent to continuous, the dynamics of fiber inflation change. What continuity changes is the rhythm at which information enters local transaction prices, not whether information arises: during closure the local price stops updating, but fundamental changes and uncertainty may still accumulate; the opening gap reflects repricing across sessions and does not automatically mean that the set of compatible states shrinks. Continuous trading shortens the interval between concentrated updates and increases the mutual citation of readings—whether new information improves identification depends on whether arbitrage, valuation, data synchronization, and risk controls adjust in step; the inflationary pressure discussed in this chapter arises from the mismatch between trading extension and calibration arrangements, not from any necessary one-way increase in uncertainty under continuity.

Boundary marking: the diagnosis of "continuous inflation" in the continuous mapping chain focuses on the disappearance of the closing moment. Prices no longer have a concentrated moment of release and calibration. But this does not mean that the quality of price signals necessarily deteriorates. Research by CME points out that futures markets have long already assumed the role of around-the-clock stock-trading platforms, with institutional investors using futures during nonstandard hours to hedge risk (CME Group, 2025). This itself shows that continuity can also serve risk management. The value of the Fiber Framework is not to assert that "continuity = bad," but to ask: once price discovery changes from intermittent to continuous, has the calibration mechanism been reconstructed at the same time? At present, the "backstage" institutions, cross-time-zone settlement procedures, data timeliness, and the coordination of regulatory rules, have not yet caught up with the speed of "frontstage" trading. This is the structural source of fiber inflation.

Input-composition marking: the continuous mapping chain makes information enter continuously, but continuity itself conceals the question of whether what is continuously entering is independent information or merely updates to existing readings. The boundary between independent components and recursive components in the partial input network becomes blurred under continuity.

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§5 The Global Financial Cycle: Synchronized Pricing under Common Driving Factors

§5.1 The Framework of the Global Financial Cycle

The "global financial cycle" framework of Miranda-Agrippino and Rey provides a macro explanation for synchronized pricing across markets (Miranda-Agrippino & Rey, 2020). Their central finding is that there exists a common global asset-price factor highly correlated with the Cboe Volatility Index (VIX), and that this factor

explains a large share of the variance in global risk-asset returns. This factor drives not only asset prices but also aggregate capital flows, so that inflows and outflows fluctuate synchronously across the globe.

Within the Fiber Framework, this means that global risk appetite (T) simultaneously drives the prices of multiple markets ($r_1, r_2, \dots, r_n$). The correlation among these price readings is not a causal chain in which "market A influences market B," but a structure in which "market A and market B are both driven by the same hidden variable." Auditors observe synchronous fluctuations in multiple market prices and attempt to identify "manipulation" or "abnormality" from that synchronicity. But the synchronicity itself may simply be driven by the global risk factor, not by inter-market causal transmission. Distinguishing "legitimate synchronicity" from "illicit manipulation" requires auditors to measure the global risk factor independently—something hard to accomplish from price data alone, because the factor is itself a latent variable extracted from price; identification must draw on additional information such as event timing, order behavior, and positions, and must state the model conditions it relies on.

§5.2 VIX as a Proxy Variable for Global Risk Appetite

The VIX measures the implied volatility of S&P 500 options and is widely used as a proxy variable for global risk appetite. When VIX rises, global risk-asset prices often come under synchronous pressure; when VIX falls, they often rise together. The direction and strength of this relation vary with asset class, period, and the nature of the shock, and may reverse for safe-haven assets.

Within the Fiber Framework, VIX is a "meta-reading." It is not the direct quote of any single concrete asset, but an implied-volatility reading extracted from S&P 500 option quotes; the global common factor of §5.1, extracted from cross-country risk-asset returns, is highly correlated with it but is not the same object. When VIX moves, all asset prices anchored to VIX move synchronously.

But the fiber of VIX itself is non-singleton: it is an implied volatility extracted from S&P 500 option prices, and its methodology is built upon the model-free continuum-of-strikes framework of Carr and Madan (1998), engineered by Cboe Global Indices (2019) through a filtering rule for deep out-of-the-money options that excludes options once two options with consecutive strike prices are found to have zero bid prices (Carr & Madan, 1998; Cboe Global Indices, 2019).

The theoretical replication relation takes a continuum of strikes as an idealization; actual calculation uses discrete option quotes screened under the same rule, and deteriorating quote quality can exclude previously included options whose bid prices have fallen to zero, changing the set of selected options and their weights (Cboe Global Indices, 2019); this book accordingly infers that the composition of the selected option set drifts significantly at the monthly, daily, and even minute scales, and that the fiber of VIX inflates sharply accordingly. These two steps are a diagnostic extrapolation made by this book on the basis of the two methodological references above, not an original conclusion of Carr and Madan or of Cboe Global Indices. For the "performativity" by which early option-pricing models such as Black-Scholes themselves shaped market practice, see the analysis of MacKenzie in the next chapter, §4. The "true state" of global risk appetite still cannot be directly observed.

It is necessary to distinguish the boundary between VIX as a "meta-reading" and VIX as a "pseudo-small fiber." VIX as a meta-reading is not itself pseudo-calibration: it is an implied volatility extracted from option prices, and its methodological limitations, model dependence and assumption sensitivity, are public information. But once VIX is unreservedly used as the proxy for "global risk appetite" and its model dependence is systematically

ignored by users, the "meta-reading" slides into a "pseudo-small fiber": a precise number conceals the fragility of the assumptions.

§5.3 Fiber Diagnosis: The Inflation of Hidden Variables

The global financial cycle reveals a deeper mechanism of fiber inflation: when multiple readings fluctuate synchronously, the synchronicity itself may be misread as evidence that "market A is influencing market B," thereby generating erroneous causal narratives, "the plunge in market A dragged down market B." In reality, both may be driven by the same hidden variable. The diagnostic value of the Fiber Framework lies in reminding auditors that synchronicity is not causality. The existence of a common driving factor leaves the causal attribution based on a single set of synchronized prices under-identified, because price contains a common factor that cannot be decomposed into the contributions of individual markets.

Input-composition marking: the global financial cycle reveals a synchronization mechanism distinct from recursive citation. When multiple markets are jointly driven by the same hidden variable, the idiosyncratic share of each market's price tends to decline. A common component does not thereby become a recursive one: reading feedback arises only when participants adjust their trading to other markets' prices, or when common indicators enter pricing rules.

The dual signal of the dollar as the master clock. Chapter 17, §6 has already established the analytic position of the dollar as the current master clock of the global financial system from two dimensions, historical, multipolarity -> unipolarity, and empirical, a three-standard verification, and has used multi-scale data—Rogoff's centennial replacement cycle, the three dollarisation waves in the U.S.-dollar share of international debt securities (IDS) documented by Pradhan et al. (2026), and IMF COFER reserve shares—to present the master clock's signals for critical monitoring: the coexistence of a falling reserve share and a persistently high payment share; Chapter 17, §6.3 explicitly notes that whether the decline in the reserve share reflects structural slippage or a wave-like swing remains to be determined, and that these single-period observations do not constitute a trend; for the classic analysis of the dollar's global monetary role, see Eichengreen (2011). This section will not repeat the empirical validation of Chapter 17, but instead places the master clock into the formal structure of the Macro-Cognitive Accounting Framework (MCAF) for structural diagnosis.

Within the Fiber Framework, the coexistence of a falling reserve share and a persistently high payment share, as presented in Chapter 17, is a candidate critical signal requiring continuous monitoring before recursion slides from functionality toward pathology. The tension between the two, the erosion of reserve-currency function and the inertia of trading/financing-currency function, is precisely the core monitoring object of the L3 recursive-closure diagnosis.

What this chapter needs to add is the dimension of **how the master clock enters risk indicators themselves**. When VIX spikes and global risk assets fall synchronously, this phenomenon is not only the manifestation of the global financial cycle in §5.1; it is also an effect of master-clock recursion. The implied volatility measured by VIX is extracted from dollar-denominated S&P 500 options, so the latent variable "global risk appetite" itself already contains the presupposition of the dollar as the unit of account.

Du, Keerati, and Schreger (2025) reveal the layered structure of this embedding: the convenience yield of the dollar, the network effect at the monetary layer, remains positive, while the convenience yield of U.S. Treasuries relative to other advanced-economy government bonds has turned negative (across the maturity structure:

persistently negative at long maturities, and at short maturities likewise since around 2023). The "monetary privilege" and the "Treasury privilege" of the master clock are no longer synchronous. This book accordingly suggests that risk indicators built on dollar-denominated assets may pass through two layers of conditions—pricing currency and safe-asset pricing—at once; but the study does not directly test how this divergence enters VIX, and this is an inference of this book rather than their original conclusion. This layering means that the L3 recursive-closure diagnosis cannot treat the master clock as a single node, but must distinguish the differentiated degradation path of its monetary layer and its asset layer.

The formal position of the master clock. The position of the master clock within the six variables of the MCAF requires precise specification. The dollar's status as the master clock is expressed in three dimensions: as a unit of account, the benchmark currency of global trade and financial contracts; as a transmission channel, Federal Reserve monetary policy independently affects global financial conditions through the dollar channel; and as a safe asset, investors globally rush into dollar assets during crisis, producing a "convenience yield" (Krishnamurthy & Vissing-Jorgensen, 2012 is the pioneering estimate of convenience yield; Del Negro, Giannone, Giannoni & Tambalotti, 2019 extend the analysis to cross-country evidence on the trend of global real interest rates). These three dimensions occupy different positions within the six variables of the MCAF.

As a unit of account, the master clock is one component of $\|L\|$: the legitimacy base of the global financial system is partly built upon the institutional consensus that the dollar serves as the benchmark currency. As a transmission channel, the master clock is a modulation factor of Λ: the global transmission network of the dollar constitutes the infrastructural layer of financial information flow. As a safe asset, the master clock serves as the anchor of the recursive closed loop: the convenience yield of dollar assets offers an entry point for analyzing the mutual reinforcement of safe-asset demand, liquidity, and network effects; when such reinforcement crosses the functional range into degeneration requires separate diagnosis—the convenience yield itself is not degeneration.

The master clock is not a seventh variable independent of the six. It is a **field-specific modulation factor** entering certain of the six variables. In different analytic settings it acts through different channels of the six variables: as the central node in the L3 recursive-closure diagnosis, as the common factor in the analysis of the global financial cycle, and as the benchmark reference in the framework of narrative arbitrage. The master clock does not appear directly in the meta-formula because the meta-formula is a cross-domain general framework. The master clock is a specific instantiation in the financial domain. It has meaning only when there exist multiple mutually referential mapping systems, and one of them is taken as the benchmark by the others.

The historical periodization of the master clock. The dollar as master clock is not something inherent in history. It formed step by step along the time axis of the degeneration of anchoring mechanisms, and the periodization itself is a diagnosis of the time axis of fiber inflation. Under the classical gold standard (1870s-1914), sterling served as the regional master clock of the gold-standard club, Europe, the United States, and Japan. Chapter 17, §6 has already argued that at the same time silver-standard China formed a parallel monetary pole; a "global unipolar master clock" did not exist before Bretton Woods. Within one gold-standard pole, central banks held gold in distributed form, the master-clock identity was substitutable, the external anchor was jointly maintained by multiple parties, and recursive depth was shallow. The two world wars and the monetary reconstruction between them (1914-1944) experienced oscillation of the dominant-currency position between sterling and the dollar, as well as repeated shifts in anchoring form, gold standard -> gold-exchange standard -> systemic collapse. The form of the anchor itself began to drift.

The Bretton Woods system (1944-1971) established a centralized architecture of the dollar pegged to gold: all participating currencies were pegged to the dollar, and the U.S. government in principle assumed the obligation of converting dollars into gold at the request of foreign governments (Rogoff, 2025, ch. 1). The master clock changed from a "substitutable lead clock" into the "institutionally designated sole benchmark"; what became centralized was the redemption responsibility and the benchmark role (U.S. monetary gold held by the Treasury), not the physical concentration of other countries' gold reserves, and the Triffin dilemma was already visible at this stage.

After the Nixon shock of 1971 closed the gold window, the external asset-redemption anchor disappeared—"anchorlessness" here means specifically the absence of an external redemption commitment, not that the currency's value is free of all other economic and institutional constraints. What sustained it was no longer a redemption commitment into gold, but three self-reinforcing loops, supplemented by the policy commitment to maintain purchasing power: the natural monopoly produced by network effects in international monetary circulation, Rogoff (2025, ch. 1) notes that about 90% of foreign-exchange trades currently involve the dollar, and that exchanging two non-dollar currencies through the dollar is often cheaper than trading them directly, even for two countries adjacent to the euro area; the depth and liquidity of the U.S. Treasury market; and the anchoring of credibility by institutional commitment. Rogoff (2025, ch. 11) points out that Bretton Woods collapsed in no small part because the United States, at the center of the system, failed to hold up its end of the bargain—the commitment to maintain stable prices; thereafter U.S. inflation accelerated in the late 1960s and deteriorated sharply in the 1970s, with severe global repercussions.

Chapter 4 of this book had already discussed this degeneration from the fiber standpoint (after the gold window closed in 1971, the ultimate calibration point disappeared, and the set of states compatible with the value of money had to be reanalyzed under new institutional constraints). What this section does is advance this discussion to the structural diagnosis of the master clock: fiber being non-singleton did not first emerge after 1971. Rather, in the stepwise degeneration of the anchoring mechanism, distributed gold holdings -> concentrated gold holdings -> anchorlessness, the rate of fiber inflation underwent critical acceleration. It is the **synchronized rise in the master clock's recursive depth and fiber-inflation rate**.

The quadrant position of the master clock. On top of temporal periodization, the master clock can be decomposed typologically along two orthogonal dimensions, whether there is an external anchor (anchored/unanchored), and who maintains the master clock (distributed/centralized). Crossing these two dimensions generates four historical quadrants: the classical gold standard (distributed + anchored, describing only the gold-standard pole; silver-standard China at the same time formed a parallel "distributed + anchored [silver anchor]" system); Bretton Woods (centralized + anchored); fiat-dollar order (centralized + unanchored); and cryptocurrency (distributed + unanchored; see Chapter 6, §3.1). It must be noted that "distributed" means different things in different quadrants: in the classical gold standard it refers to central banks holding gold in distributed form, whereas for cryptocurrency it refers to the decentralization of the verification and maintenance architecture.

The evolution among these four quadrants is not linear progress. The move from the classical gold standard to Bretton Woods was a "centralization of the anchor," with central banks concentrating gold holdings. The move from Bretton Woods to the fiat-dollar order was the "disappearance of the anchor," the 1971 Nixon shock. The move from the fiat-dollar order to cryptocurrency is "decentralization of structure while the absence of an anchor remains." The centralized/decentralized dimension and the anchored/unanchored dimension follow different

trajectories and cannot be reduced to a single sequence of superiority or inferiority. The current dollar occupies the quadrant "centralized + unanchored," where the external anchor has degenerated into a self-reinforcing loop of institutional commitment and network effects. This quadrant sharpens the recursive-closure diagnosis in §5.3.

The cryptocurrency entry refers only to decentralized systems without an external redemption commitment and does not cover all crypto-assets; the diagnosis of Chapter 6, §3.1 is that the crypto market has cyclical shocks but no direction of calibration—tokens lacking redemptionable asset support have no underlying asset to anchor the direction of return after a crash, and the price's reversion to "community consensus" constitutes another self-reference.

This chapter adds to that diagnosis: a distributed architecture removes the dependence on any single central node but does not alter this self-referential structure—"distributed + unanchored" is not superior to "centralized + unanchored"; it merely has a different mode of failure. Chapter 22 will draw on this quadrant diagnosis in comparing the calibration of four financial scenarios.

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§6 Market Fragmentation: The Physical Dilemma of Data Integration

§6.1 Evidence of the Scale of Fragmentation

The 2025 market analysis of ION Markets provides evidence of the scale of market fragmentation. Roughly 50% of U.S. equity trading volume is executed over the counter, while in Europe about 70% remains executed on exchanges. Even within the on-exchange space fragmentation is considerable; Baker (2025), citing an Oxera report, notes that traded volume in the UK is split between lit order books, auctions, dark venues, and Systematic Internalizers. A full first-hand record of this micro picture of fragmentation appears in Patterson's *Dark Pools* (2012; Chinese edition, China Machine Press, 2015): orders flow to the four public exchanges, NYSE, Nasdaq, Direct Edge, and BATS, each with multiple sub-markets beneath them; by his account, trading orders emanate from more than fifty venues, about one-third of them taking place in the dark.

Within the Fiber Framework, fragmentation means that the mapping chain changes from a "single reading" into "multiple dispersed readings." In structures where trading was relatively concentrated, the quotes of the leading venues could readily serve as the market reference; fragmentation raises the burden of reconstructing quotes and execution conditions across venues—the same stock now trades simultaneously on more than a dozen venues, each with its own quote, trading volume, and participant structure. The auditor must integrate data from all of these venues in order to reconstruct the complete picture of "what the true price of this stock is."

§6.2 The Dual Effect of Fragmentation on Liquidity

Aliyev et al. find that algorithmic trading and market fragmentation have a dual effect on liquidity in equity markets (Aliyev, Aquilina, Rzayev & Zhu, 2024): they tend to reduce average spreads (improving liquidity), but increase the skewness of spreads (so that liquidity evaporates sharply during periods of stress). This means that during calm periods fragmentation makes the market look more efficient, because spreads are smaller; rising skewness signals the deterioration of liquidity in the tail—the improvement of trading costs in calm times can coexist with fragility under stress, though this is not sufficient to establish a synchronized withdrawal of

liquidity across all venues.

Within the Fiber Framework, this is another form of pseudo-small fiber: the small bid-ask spread in calm periods creates the sense of certainty that "market liquidity is abundant," but the uncertainty structure behind that reading, the extreme deterioration of liquidity's tail behavior in stressed periods, is invisible to participants. The reading appears precise, spreads are very narrow, but the fiber is in fact large, because the tail behavior of liquidity is extreme.

§6.3 Fiber Diagnosis: The Integration Dilemma of Dispersed Readings

The fiber significance of fragmentation is this: given a set of quotes dispersed across multiple venues, the integration of those quotes into one "national best bid and offer" is itself a lossy mapping. Under the U.S. Regulation NMS framework, the NBBO, National Best Bid and Offer, is precisely such an integrated reading: it compresses the quotes from more than a dozen venues into one "best bid" and one "best offer." But the fiber of the NBBO is non-singleton: different venues update at different speeds, direct data feeds versus the Securities Information Processor (SIP), and at the instant of integration the quote may already be stale. The NBBO aggregates the best quotes under specific rules, but it does not contain the full order book, total trade sizes, or every participant's execution conditions—its lossiness means it cannot be treated as the one and only price, unaffected by timing, size, and execution constraints; this does not mean the quotes are false or that the prices cannot be executed. Fragmentation thereby deconstructs the exclusivity of the "single true price" presupposition: what exist are multiple dispersed readings and, derived from them, a "best estimate" whose fiber remains non-singleton.

In 2025, the SEC began a full re-evaluation of the pseudo-calibration locked in by the NBBO regime. Patterson's *Dark Pools* (2012) records the institutional background of Regulation NMS: in 2005 the SEC issued the Regulation NMS (fully effective in 2007) in an attempt to glue fragmented electronic trading markets into one interconnected network. On that account, its core requirement was that stock orders must be routed to the venue with the best price, and that price information be shared through the SIP mechanism—a paraphrase preserved as institutional-history record; the rule itself is a trade-through prohibition on protected quotes, with exceptions. But Patterson also records the "vast tangle of complications" spawned by Reg NMS: every trading venue had to constantly monitor the price of a stock (or hundreds or thousands of stocks) on every trading venue, all the time, and any price change could lead to an order being rejected, rerouted, or kicked to the back of the queue.

This is precisely the institutional legacy that later prompted the SEC to launch its re-evaluation. On September 18, the SEC convened a roundtable devoted specifically to discussing the market impact of Rule 611, the Order Protection Rule—three panels discussed, respectively, participants' practical experience, the role of the rule in the current market structure, and future directions of reform. In his opening statement, SEC Chair Atkins criticized Rule 611, stating that "the fragmentation that has followed Reg NMS is now obvious" and that "a rule intended to strengthen liquidity has instead splintered it among an unprecedented number of venues" (Atkins, 2025), and further noted that the economic analysis in the original approval had failed to confront the risk of fragmentation and its adverse implications for competition. As Lexology (2025-12-18) recounts, Atkins stated at the roundtable that Rule 611 has produced market fragmentation, dispersed liquidity, and diminished transparency, and called for a comprehensive re-evaluation of the rule and related provisions. Earlier, in the SEC press release announcing the roundtable (July 2025), Atkins had said that Reg NMS and Rule 611 "have not served investors or broker-dealers well, given the market distortion and resulting gamesmanship by those

that seek to take advantage of the Reg NMS structure" (U.S. Securities and Exchange Commission, 2025b); before the roundtable, industry media had already reported this re-evaluation trend (Lyudvig, 2025; see also Delta Strategy Group, 2025). At the same time, the SEC advanced market-structure reform in multiple stages—in November 2025, price-tiered dynamic round-lot definitions took effect (e.g., 100 shares retained for stocks under $250 and reduced to 40 shares for the $250-$999 tier), bringing into the consolidated data some quotes previously classified as odd lots and excluded from the NBBO; adjusting the round-lot boundary and separately disclosing odd-lot information are related but distinct reforms (Feldman, 2026).

In fiber language, the SEC's re-evaluation shows that the consequences of the existing price-protection arrangements remain contested—a design once regarded as "protecting investors" has had its fragmentation consequences placed on the regulators' re-evaluation agenda; this provides an institutional-history echo for "institutionally locked pseudo-calibration," but does not amount to official adoption of that diagnosis. But the re-evaluation runs on τ_g (the governance clock, Chapter 3, §3.2): from roundtable, to rule revision, to full implementation, the timescale is measured in years, whereas fragmentation runs on τ_c (the technological-capital clock). Pseudo-small fiber, in its cross-venue spread, also exhibits an adversarial dynamic: adversarial-learning experiments show that even when a detector retains perfect detection against the manipulative strategy of the current generation, the generator can still evolve a new generation of evasive strategies—manipulation effectiveness falls from a baseline of 1.00 to 0.22 over successive generations in exchange for the capacity to evade detection—within the experimental generations, detection and evasion settled into a mutually constraining tug-of-war with no convergence observed (Wang & Wellman, 2020). The monitor-evasion adversarial loop leaves the "liquidity reading under fragmentation" facing a dual source of inflation: cross-venue delay and adversarial evolution.

In the same period, Aquilina et al. (2025) built and tested a recurrent neural network (RNN) tool for warning of market dysfunction, mitigating the black-box nature of the RNN through data-driven time-varying input weights, which makes its decision process transparent (Aquilina, Araujo, Gelos, Park & Perez-Cruz, 2025, BIS Working Paper No. 1291)—this self-reflective discussion by Aquilina et al. and the SEC's reconsideration of Rule 611 converge independently along the recursive dimension: whether one uses institutional tools (Reg NMS / Rule 611) or regulatory-side AI tools (warning RNNs), in a fragmented and high-dimensional nonlinear market one cannot penetrate fiber all the way to a single true state.

Fragmentation may have a hidden fiber effect on speed-limit regulation. If participants can observe throttling feedback, or infer from identifiable order behavior that other accounts are near a rate threshold, the speed-limit arrangement itself may become a strategic input: the fact that the order rate on one venue is approaching the threshold leaks probabilistic information that "another high-frequency participant is close to the limit on this venue," which can be used for venue-jumping arbitrage; fragmentation increases the possibility of migrating execution across venues. Whether this mechanism holds depends on information visibility, account rules, and cross-venue migration conditions, and cannot be inferred from the number of venues alone—it is a candidate fiber dimension requiring audit, not an established fact.

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§7 High-Frequency Trading: From Market Quality to Transmission into the Real Economy

§7.1 The Dual Effect of High-Frequency Trading on the Cost of Capital

The study released by the BIS in September 2025 examines how high-frequency trading affects the real economy through the cost of capital. Aquilina et al., using Nasdaq colocation and latency-improvement upgrades as a natural experiment, find that high-frequency trading (HFT) has a dual effect on the cost of capital (Aquilina, Ibikunle, Rzayev & Wang, 2025): for low-beta stocks, HFT raises financing costs by amplifying systematic risk. The concrete mechanism is that HFT's correlated trading strategies make these stocks more responsive to market-wide information, thereby amplifying their systematic risk. For the most liquid stocks, HFT lowers the cost of capital by reducing the liquidity premium. The same effect appears in the unfragmented Hong Kong market, proving that it is not driven by fragmentation itself.

Within the Fiber Framework, this means that HFT not only changes market microstructure, but transmits its effects into the real economy through the cost-of-capital channel. The mapping chain is no longer one-way, "true state -> price." It becomes "price-formation mechanism -> cost of capital -> corporate investment -> true state," a complete feedback loop. HFT increases the sensitivity of prices to systematic risk factors, so that the price movements of low-beta stocks follow overall market movements more closely. The hidden subspace continuously inflates along the dimension of "idiosyncratic information": HFT makes some stocks more sensitive to market-wide information and raises their systematic-risk exposure, changing the relative weights of the common and idiosyncratic components of price movement; what this chapter attends to is whether the change increases the difficulty of distinguishing firm-specific factors from common shocks within a single price.

§7.2 Fiber Diagnosis: The Completion of the Feedback Loop

High-frequency trading pushes the feedback loop of the mapping chain into a new range of intensity. The feedback relation whereby prices influence corporate investment through the cost of capital was not first created by HFT; what HFT changes is the risk exposure, liquidity premium, and strength of influence within that pre-existing channel—changes in market microstructure are thereby transmitted into corporate decisions: the mechanism of price formation in turn shapes the true state. This is not simply an enhanced version of "price discovery"; it is a new mechanism of "price shaping." Price not only reflects the real; through the cost-of-capital channel it changes the real.

Once the mapping chain becomes closed, the audit faces a dual task: identifying the set of states compatible with the current price (the inverse image $\pi^{-1}(r)$), and assessing the paths through which that price shapes future states via financing and investment (dynamic feedback analysis)—the former is inverse-image identification, the latter dynamic analysis, and the two cannot be merged into a single $\pi^{-1}(r)$ without redefining the state space; if the state is defined as a complete time path, the corresponding observation mapping must be specified separately. The inflation of fiber is therefore double: one must both trace backward into the past and project forward into the future. This double tracing presupposes that the causal chain "price shapes reality" is itself reversible at the moment of tracing, but under the spatial coexistence of algorithms this presupposition itself faces an attribution challenge.

In cross-market settings where algorithms coexist, the feedback loop also raises problems of legal attribution: even in single-market simulation, reinforcement learning algorithms can autonomously sustain collusion without any agreement, communication, or intention (Dou, Goldstein & Ji, 2025). Given observations of informed

algorithms sustainably earning supra-competitive profits, $\pi^{-1}(r)$ cannot be uniquely attributed to any of the three true states of "agreement-type collusion / emergent collusion / efficient pricing." The classic antitrust framework, with agreement as a constitutive element, faces attribution difficulties before strategy convergence formed by autonomous learning—the economic notion of a coordinated outcome and the legal finding of collusion are not the same, and the specific elements and remedies vary by jurisdiction (Dou et al. describe such collusion as an autonomous outcome sustained "without agreement, communication, or intent").

But this collusion is conditional rather than universal: Keppo et al. (2026) find that patience heterogeneity, asymmetric data access, cross-algorithm heterogeneity, and a larger number of competing agents all weaken the collusive premium in their model; model-size differences do not, and instead stabilize leader-follower dynamics. The conclusion of attribution difficulty therefore cannot be generalized into "cross-market algorithmic coexistence necessarily colludes"—collusion is highly sensitive to deployment heterogeneity, and that fragility is itself a diagnostic clue.

Boundary marking: the "dual effect" found by the BIS reveals an important asymmetry. The effect of HFT on the cost of capital is not uniform: for low-beta stocks, HFT raises financing costs by amplifying systematic risk; for the most liquid stocks, HFT lowers the cost of capital by reducing the liquidity premium. Beta and liquidity are different dimensions, and the two groups can overlap; the average effect across the sample remains a higher cost of capital. This means that the systemic consequence of HFT is not a binary judgment of "good" or "bad," but a **distributional** one: the financing costs borne by different stocks move in opposite directions. The regulatory implication of this boundary is that uniform rate-type controls, such as rules that tier high-frequency-trading designations by the combined number of order submissions and cancellations per second, may suppress at the same time the liquidity-supply function of HFT in high-liquidity stocks—designation thresholds, monitoring standards, and behavioral caps are different kinds of institutions, and assessing their impact on liquidity requires distinguishing the type of rule first (Shanghai Stock Exchange, 2025). Precision regulation requires differentiated constraints for different tiers of liquidity.

§7.3 Audit Tools as a New Layer of the Mapping Chain

Once audit tools themselves become algorithmic, a structural change occurs: they begin producing new readings, and those readings are in turn taken by market participants as inputs for optimizing their own behavior.

This transformation is marked by four jumps. The Small Order Execution System (SOES, implemented in 1985 and made mandatory after the 1987 crash) automated the execution of small orders; it did not produce a new reading, but merely removed human decision from the execution loop. The National Best Bid and Offer (NBBO, issued in 2005 and effective in 2007), through the Securities Information Processor (SIP), aggregated quotes across multiple exchanges and produced the synthetic reading of "the national best bid and offer," a reading independent of any single venue. This reading is constrained by integration latency and cross-venue asynchrony (§6.3), but once it entered the market network it became an input for pricing models and trading algorithms. The Market Information Data Analytics System (MIDAS, 2013) reconstructed the full market message flow in real time and generated readings of market-health indicators; those readings in turn were used by traders to design signal patterns capable of evading monitoring. The Consolidated Audit Trail (CAT, from 2020 onward) reconstructs the full trading chain after the fact, and the results of that reconstruction, which patterns are flagged, which are not, become a probabilistic source of information for market participants assessing the likelihood that regulators will grasp certain behaviors.

NBBO, MIDAS, and CAT—three of these jumps—share one common structural feature: audit tools are not observational devices external to the market; SOES, as a precursor of execution automation, prepared the ground but does not belong to this sequence. Recording, aggregating, reconstructing, and restoring themselves produce new readings, and those readings are embedded in the mapping chain as intermediate nodes parallel to trading algorithms and pricing models. The readings of audit tools themselves also have fiber, and are subject to the same general conditions diagnosed in the preceding sections, non-singleton fiber, pseudo-small fiber, hidden subspace, recursive closed loop, and so on. These difficulties can therefore arise a second time at the audit layer. This is not a passive concession of the form "speed has become too great, so hand it over to algorithms." Rather, the audit tool itself has entered the very mapping chain it is trying to audit.

The problem here does not lie in "speed" itself, but in the mismatch of action cycles: order placement and cancellation proceed on the timescale of microseconds and milliseconds, while investigation, hearing, and adjudication proceed on the timescale of weeks, months, and years. Algorithmization attempts to compress audit onto a commensurable scale. SOES began with automation at the execution side in 1985; by CAT this had advanced to post hoc full-chain reconstruction from 2020 onward. But the cost of that commensurability is that the audit tool itself becomes one layer of the mapping chain: its readings in turn become optimization inputs for the algorithms it seeks to audit. **The result is not that "regulation has caught up with the market," but that the difficulty of "what humans cannot see clearly" has been transformed into the difficulty of "what algorithms cannot measure accurately."** The general conditions diagnosed in the preceding sections reappear at the audit layer in the form of new readings.

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§8 Conclusion: Fiber Diagnosis in Continuous Markets

When price discovery changes from a single market session into a global 24-hour continuous relay, the structure of the mapping chain undergoes a fundamental change. This chapter provides an integrated diagnosis of that transformation by using the four-layer progressive structure of the Fiber Framework, L0-L3, the diagnostic tool established in Chapter 4, §7; the financial version of the concept of "partial input network" derives from Chapter 17, §1.2.

L0 (mapping-chain structure): the mapping chain changes from the simplified-chain structure of "true state -> local reading" into the reticular structure of "true state -> globally dispersed readings -> mutual calibration -> emergent price." The night-session system connects local mapping chains into the global continuous network. Cross-border ETFs and ADRs perform proxy pricing in closed markets. Twenty-four-hour futures trading disperses price discovery across multiple time zones. The length of the mapping chain is no longer measured only by the number of layers of information processing, but also by the continuity of the time window and the degree of geographic dispersion.

L1 (optimization dynamics): the precondition of pricing power belongs to "the market that happened to be open when the information shock occurred," but access rights alone do not amount to dominance. The institutional design of global continuous trading is in reality a redistribution of access rights to the mapping chain. High-frequency trading further changes optimization dynamics: the optimization targets of algorithms, minimizing inventory risk and maximizing arbitrage profit, make prices reflect systematic risk factors more strongly and firm-specific information less strongly. The hidden subspace continuously inflates along the dimension of

"idiosyncratic information."

L2 (pseudo-calibration): the proxy pricing of cross-border ETFs and ADRs creates a particular form of pseudo-small fiber. ETF prices during the closure of the underlying market provide investors with a precise reading, but the uncertainty structure behind that reading, arbitrage constraints and liquidity shocks, is invisible to investors. Market fragmentation makes the NBBO into another pseudo-small fiber: it integrates the quotes of more than a dozen venues into one "best price," but the latency and asynchrony of the integration process hide great uncertainty behind that precise reading.

L3 (multi-mapping recursion): the mutual calibration among multiple markets forms a global-scale recursive closed loop. The ADR market and the home market each become the other's "other clock," mutually calibrating one another. The global financial cycle causes multiple markets to move synchronously without direct causal interaction. That synchronicity is itself driven by one hidden variable, global risk appetite, rather than by causal transmission from one market to another. The recursive closed loop means that, given a price reading in any one market, the true state cannot be uniquely determined, because the reading has itself already become a function of other market readings.

These changes cannot be attributed solely to regulatory failure or investor judgment; they must be examined jointly through information access, arbitrage conditions, data integration, and feedback structure. In the structural reorganization of the partial input network on a global scale, the geographic sources of independent information are redistributed, and mutual citation among readings intensifies. The task of the Fiber Framework is not to "solve" this evolution, but to mark it honestly: once the subject of price discovery changes from "a single market" into "a global network," the length of the mapping chain, access rights, the forms of pseudo-small fiber, recursive closed loops, and the direction of feedback all change fundamentally. The very concept of "audit"—tracing back from a reading to the true state—has to be redefined. In global continuous markets, no single reading suffices to uniquely restore the true state—the global price keeps receiving independent information, yet the mutual citation of readings means that no single market reading can by itself identify the states and mechanisms behind it; the "global price" that emerges from this process is one whose fiber, within any foreseeable calibration arrangement, does not degenerate into a single point.

This chapter completes the "spatialization" of the financial mapping chain: the diagnosis of L0-L3 at two scales, within the single market (Chapter 17) and within the global continuous market (this chapter). Chapter 20 will turn to how pricing models and institutional ideas filter information (the design of the mapping function and its epistemic premises), and Chapter 21 will further examine products and practitioners (the micro carriers).

The next chapter, Chapter 20, will shift the diagnosis from market structure to the epistemic level: pricing models themselves are not ideologically neutral. Shareholder value maximization, the Efficient Market Hypothesis, and the performativity of financial models together constitute the built-in filter of the mapping function, systematically pushing particular dimensions into the hidden subspace. This is not a problem of market speed or continuity, but a problem of the design of the mapping function π itself.

Terminology Correspondence Table

The correspondences between the global continuous market structure diagnosed in this chapter, the Fiber Framework, and the institutional and clock implications are as follows:

Diagnostic object in Chapter 19	Correspondence in the Fiber Framework	Institutional and clock implications
The night-session system connects into the global pricing network	L0 mapping-chain structure—reconnecting a broken mapping chain	Geographic expansion of Λ (media conductivity coefficient)
Cross-border ETFs engage in proxy pricing during the closure of the underlying market	Pseudo-small fiber (L2)—the pseudo-small fiber of proxy pricing	Trading, valuation, and arbitrage conditions involve time lags; not equivalent to failure of the governance clock
Extended trading hours (including 24/5 and 24/7 arrangements)	L0—the mapping chain changes from intermittent to continuous	The mismatch between calibration arrangements and trading extension becomes the source of inflation
The global financial cycle	Inflation of hidden variables—synchronicity does not equal causality	Common shocks may lower the relative share of local idiosyncratic information; not automatically recursive
The NBBO integrates dispersed quotes	Pseudo-small fiber (L2)—a precise reading conceals cross-venue uncertainty	—
The SEC's re-evaluation of Rule 611	An institutional-history echo of institutionally locked pseudo-calibration and regulatory re-evaluation	Rule adjustment proceeds along the governance cycle, measured in years, and may affect the institutional legitimacy base
HFT -> cost of capital -> corporate investment	L3 recursive closed loop—the price-formation mechanism feeds back into the true state	The intensity and distribution of the pre-existing feedback channel enter a new range

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